

Optimizing Spatial Feature Extraction: Evaluation of the `pixels_per_cell` Parameter in HOG for FER Systems[★]

Adam Karol Drażyk¹, Marta Maja Załuska²

¹*Faculty of Applied Mathematics, Silesian University of Technology, Kaszubska 23, 44100 Gliwice, POLAND*

Abstract

Despite the dominance of deep learning networks in the field of Facial Emotion Recognition, classical feature extraction algorithms such as HOG are still a highly relevant alternative for specific use cases. One of the most underestimated factors of this solution is the proper selection of parameters, which can strongly affect the results. In this paper an attempt to evaluate one of the HOG parameters `pixels_per_cell` was made. For this purpose the CK+ dataset, containing 981 images of facial expressions, divided among seven different emotion classes, was utilized. The data was subject to preprocessing using the HOG algorithm with different `pixels_per_cell` values. To compare the effect of those values on classification performance while preventing data leakage, a grouped cross-validation strategy was applied to evaluate a Random Forest model and a referential Support Vector Classification model. After comparing the F1-Score values and computing times, the values of (4, 4), (6, 6), (8, 8) were selected as best-performing, with (8,8) being proposed as the best compromise between classification performance and computing time.

Keywords

FER, HOG, facial emotion recognition, histogram of oriented gradients, parameter, random forest

1. Introduction

Nowadays, Facial Emotion Recognition (FER) is one of the most important areas of computer vision, primarily due to the growing demand for the development of human-machine interaction [1]. In recent years, deep learning networks have become dominant in this field, achieving high predictive performance [2].

The main disadvantage of deep learning models is the fact that they require massive training datasets and high computational power, and they are characterized by low interpretability of the decision-making process [3]. For this reason, machine learning methods based on classical feature extraction still constitute a viable alternative [4].

One of the more frequently used tools in image analysis is the HOG (Histogram of Oriented Gradients) algorithm. The algorithm was initially used in human silhouette detection [5], but currently, it is increasingly encountered in the analysis of face recognition [6]. The effectiveness of the HOG algorithm in FER mainly depends on the selection of parameters determining the specific division of the analyzed image, in particular on the `pixels_per_cell` parameter,

SSI, project 2025/2026

✉ ad315968@student.polsl.pl (A. K. Drażyk); mz315950@student.polsl.pl (M. M. Załuska)



© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).



CEUR Workshop Proceedings (CEUR-WS.org)

which is responsible for the cell size. Most publications set this parameter heuristically or use default values, even though such settings may degrade the model's ability to capture details [7].

In this paper, an attempt to fill the research gap was made, through a statistical evaluation of the impact of the `pixels_per_cell` parameter of the HOG algorithm on classification performance. To demonstrate the universality of the formulated theses, the study was conducted on the Random Forest algorithm and the Support Vector Classification reference model.

2. Methodology

2.1. Feature Extraction

The dataset of 981 facial images was analyzed, obtained from the CK+ database [8], divided into 7 emotion classes. Initially, each image was subjected to normalization, and subsequently to HOG feature extraction utilizing the `scikit-image` library [9]. The algorithm was run with the parameters: `orientations = 8`, `cells_per_block = (2, 2)`, `transform_sqrt = True`, `feature_vector = True`. The investigated `pixels_per_cell` parameter took values from the set $\{(2, 2), (4, 4), (6, 6), (8, 8), (12, 12), (16, 16)\}$.

According to the original article by N. Dalal and B. Triggs, *Histograms of oriented gradients for human detection*, the parameters `transform_sqrt`, `cells_per_block` utilized in this study are optimal parameters [5], while initial parameter optimization showed that for the analyzed CK+ dataset, the value `orientations = 8` provides higher metric stability than the classical value of 9. The `feature_vector` parameter set to the value of `True` ensured that the algorithm, as a result of its operation, would return a flattened one-dimensional vector instead of a result matrix. That vector became the final input to the Random Forest classifier.

The HOG feature extraction algorithm used the components G_x, G_y . They were calculated based on the function $I(x, y)$, which takes the pixel coordinates x, y as an argument and returned the pixel intensity as a value, as an integer in the range $[0, 255]$:

$$G_x(x, y) = I(x + 1, y) - I(x - 1, y), \quad (1)$$

$$G_y(x, y) = I(x, y + 1) - I(x, y - 1). \quad (2)$$

Based on the components G_x, G_y , the gradient magnitude m and its orientation θ were calculated [5]:

$$m(x, y) = \sqrt{G_x(x, y)^2 + G_y(x, y)^2}, \quad (3)$$

$$\theta(x, y) = \arctan \frac{G_y(x, y)}{G_x(x, y)}. \quad (4)$$

After calculating the values θ, m for each pixel, the entire image was divided into a grid of smaller cells, whose size is controlled by the `pixels_per_cell` parameter. For each cell, a histogram of eight orientations was created. The angle θ determines which of the eight bins a given pixel would fall into, and its magnitude m constituted the weight added to this bin.

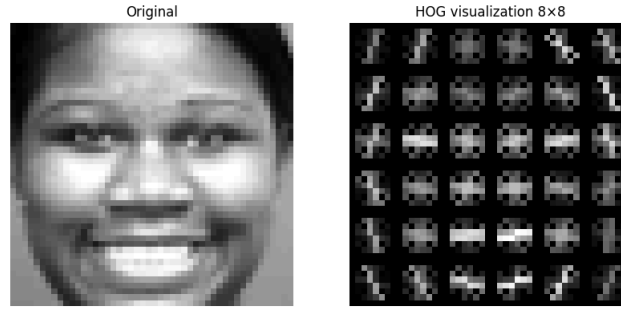


Figure 1: Visualisation of HOG extraction for `pixels_per_cell = (8, 8)`

2.2. Classification Models

As the main classifier, the Random Forest algorithm configured with a parameter `n_estimators = 100` was utilized, constructed using the `scikit-learn` package [10]. This algorithm was chosen due to its ability to handle multidimensional feature vectors, such as flattened HOG histograms. A single node of the decision trees within the forest was split based on the Gini Impurity measure, determining the probability of incorrect classification of a random sample [11]:

$$I_G = 1 - \sum_{i=1}^c p_i^2, \quad (5)$$

where c is the number of classes (in our case $c = 7$), and p_i was the probability of the occurrence of a sample of a given class at the split node.

In order to objectively evaluate the performance of the Random Forest, the Support Vector Classification (SVC) reference model, also implemented via the `scikit-learn` package [10], was introduced to the study. Due to the high dimensionality of the data, an algorithm with a linear kernel (`kernel = 'linear'`) was applied. The choice of the SVC model was dictated by its proven effectiveness when combined with the HOG feature extraction algorithm [5].

2.3. Evaluation Metrics

Because the dataset consists of multiple expressions derived from the same person, standard cross-validation could result in the model's learning to recognize specific facial features rather than generalized emotions. To prevent this, a 5-fold Group Cross-Validation (`GroupKFold`) strategy was implemented.

This approach utilizes subject identifiers as a grouping factor, ensuring that all images belonging to a single person appear either entirely in the training set or entirely in the validation set. This guarantees that the model's predictive capability is evaluated on completely unseen individuals. Furthermore, to ensure full reproducibility of the experiments, the Random Forest classifier was initialized with a fixed random seed (`random_state=7`).

To evaluate the performance of the trained classifiers, standard metrics were used during the cross-validation process: accuracy ACC, precision PPV, recall TPR [12], and due to the multiclass nature of the data, the F1-Score F_1 was adopted as the main comparative measure. This metric constitutes the harmonic mean of precision and recall:

$$F_1 = 2 \cdot \frac{\text{PPV} \cdot \text{TPR}}{\text{PPV} + \text{TPR}}. \quad (6)$$

All metrics were calculated in the macro variant, which means that each of the classifier's evaluation measures was calculated independently for each of the seven emotions, and then their arithmetic mean was calculated, which constituted the final metric. Such an approach prevented favoring majority classes and provided an objective evaluation of the algorithm's stability across the entire dataset.

3. Experiments

3.1. Dataset

The CK+ dataset (The Extended Cohn-Kanade Dataset) [8] was used in this study. It originally consisted of 593 unique video sequences, 327 of which have been classified with one of the 7 emotions: Anger, Contempt, Disgust, Fear, Happy, Sadness or Surprise. Each sequence started with a neutral expression and ended with a peak expression.

To adapt the data for static image processing, three peak frames were extracted for each sequence. This yielded 981 greyscale images in 48×48 pixel resolution, all of which were then utilized in the study. The database with those images is publicly available via Kaggle [13].

Emotion	Class size
Contempt	54
Fear	75
Sadness	84
Anger	135
Disgust	177
Happy	207
Surprise	249

Table 1
Class distribution in the dataset

3.2. Classification

To compare classification results' dependency on the `pixels_per_cell` (PPC) parameter, six different PPC values were selected: (2, 2), (4, 4), (6, 6), (8, 8), (12, 12), (16, 16). For each PPC value, two classification models were implemented – a Random Forest model (RF) as well as a referential Support Vector Classification model (SVC).

As every three consecutive images in the utilized dataset were extracted from the same video sequence, all three of them depicted the same person. To address this structure, the

GroupKFold strategy was applied, using the video sequence IDs embedded in image names as the grouping factor. This practically enforced that all three peak frames of a given subject’s expression were kept within the same fold, effectively preventing data leakage.

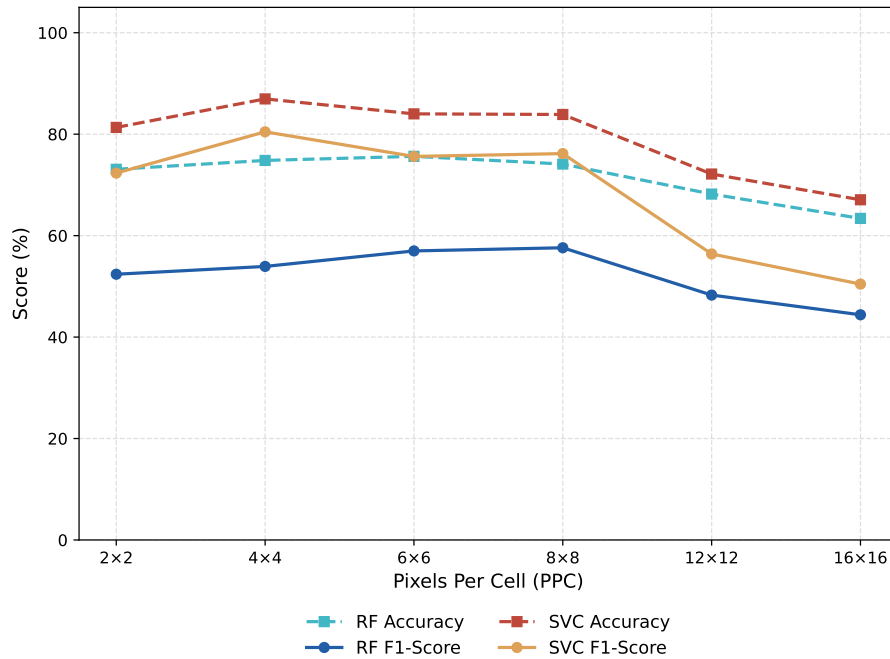


Figure 2: Classification results depending on pixels_per_cell value

pixels_per_cell	(2, 2)	(4, 4)	(6, 6)	(8, 8)	(12, 12)	(16, 16)
RF F_1 (%)	52.39	53.93	56.98	57.61	48.28	44.40
SVC F_1 (%)	72.33	80.46	75.62	76.17	56.39	50.45

Table 2

F_1 depending on pixels_per_cell value

Figure 2 illustrates how the PPC parameter influences Accuracy and F1-Score in both RF and SVC models, while Table 2 specifies the exact F_1 values. Similar trends can be observed for RF and SVC.

Considering PPC values (2, 2), (4, 4), (6, 6), (8, 8), there was a slight, but steady increase for RF, starting from $F_1 = 52.39\%$ for (2, 2) and peaking at $F_1 = 57.61\%$ for (8, 8). For SVC, a sharper increase can be observed when changing the value from (2, 2) ($F_1 = 72.33\%$) to (4, 4) ($F_1 = 80.46\%$). For subsequent PPC values, the F_1 percentage drops to $F_1 = 75.62$ for (6, 6) and $F_1 = 76.17$ for (8, 8). It can be seen that the use of PPC values greater than (8, 8) decreased all metric scores, with models yielding the worst results for (16, 16) ($F_1 = 44.40\%$ for RF and $F_1 = 50.45\%$ for SVC).

Regarding classification quality, the best performing PPC values were (4, 4), (6, 6), (8, 8). The results indicate that HOG extraction using (2, 2) generates too much visual noise, while

(12, 12) and (16, 16) blur crucial expression details, thus lowering Accuracy and F_1 [7].

To assess the optimal `pixels_per_cell` configuration for Facial Expression Recognition (FER), it is also important to recognize how each value affects the computing time in both models. Table 3 compares those statistics. The computing time measured by the program includes solely training a model and classifying data – it does not contain HOG extraction in itself.

<code>pixels_per_cell</code>	(2, 2)	(4, 4)	(6, 6)	(8, 8)	(12, 12)	(16, 16)
RF computing time (s)	0.39	0.32	0.24	0.21	0.17	0.15
SVC computing time (s)	59.85	0.41	0.09	0.04	0.02	0.01

Table 3

Computing time depending on `pixels_per_cell` value

It can be seen that the computing time in both models decreases as the PPC value increases. Taking into account only the values selected previously for (4, 4), (6, 6), (8, 8), the most time-optimal configuration for `pixels_per_cell` would be (8, 8). In comparison to (4, 4), it is almost two times faster using RF and 17 times faster using SVC. The time decrease is a natural consequence of the drastic dimensionality reduction. Although the exact performance gain depends on the internal architecture of each classifier, both models inherently benefit from a smaller feature space.

3.3. Optimal configuration

Among all tested `pixels_per_cell` values, (8, 8) emerged as the optimal configuration. Its quality was the highest among all the configurations tested when using Random Forests. For Support Vector Classification it generated third-best classification metrics; however, it computed much faster than other well-classifying values. Thus, (8, 8) is the optimal PPC value for RF and is the best compromise between time and quality for SVC.

To further examine its performance, a detailed confusion matrix was prepared, as shown in Figure 3.

Figure 3 shows that images belonging to the Happy and Surprise classes were almost always correctly classified (96% and 97%, respectively). Samples from the Disgust category were also recognized with high accuracy, achieving 90%. Images representing the Anger class yielded an accuracy of 66% and were most frequently misclassified as Disgust (22%). Instances of Contempt and Fear presented greater challenges for the model. The Contempt class scored 22% and was often confused with Anger (31%) and Happy (15%), whereas the Fear class achieved 20% and was predominantly misclassified as Happy (24%), Surprise (19%), and Disgust (16%). The most challenging category proved to be Sadness, with only 13% of its samples correctly identified, as they were overwhelmingly misclassified as Anger (50%) and Surprise (21%).

The variation in classification accuracy across different classes stems from the specific facial deformations evoked by the underlying emotions. The highest-performing categories, such as Happy, Surprise, and Disgust, correspond to emotions expressed through vivid and unique facial movements, such as a wide-open mouth or raised eyebrows. Conversely, samples from classes like Contempt, Fear, and Sadness represent emotions that involve only subtle changes, such as slight lip or eyebrow raises, creating a less pronounced overall expression[14]. Depending on

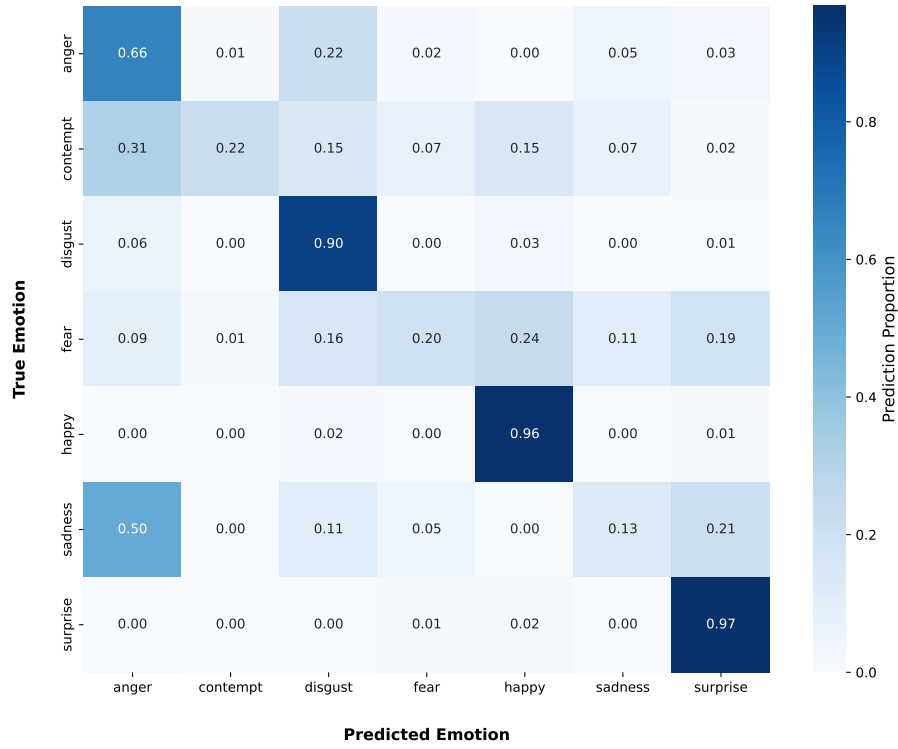


Figure 3: Confusion matrix for Random Forest (RF) and `pixels_per_cell = 8`

the amplitude of a given facial expression, the HOG extraction algorithm may smooth out the deciding details [5], leading the classifier to confuse these closely related classes.

4. Conclusion

The aim of the study was to assess the optimal value of the `pixels_per_cell` (PPC) parameter in Histogram of Oriented Gradients (HOG) extraction for Facial Expression Recognition (FER). Images displaying seven different emotions from the CK+ dataset [8] were used to test the classification quality. HOG extraction was performed on the images six times, each time with a different PPC value from the chosen set $\{(2, 2), (4, 4), (6, 6), (8, 8), (12, 12), (16, 16)\}$. Then, two classification models (Random Forest and Support Vector Classifier) were applied to the extracted gradients. For each PPC value, the F1-Score and computing time were measured for both models. Regarding classification quality, the values $(4, 4)$, $(6, 6)$, $(8, 8)$ were selected as best and after comparing computing times, the optimal value $(8, 8)$ was selected as a compromise between computing time and classifying quality.

The study could be expanded in the future to explore any factors that may influence the results. Firstly, the dataset used in the research was relatively small and the classes were unbalanced. Also, the study was only conducted on one dataset; therefore, validating those results on other datasets would be needed. It is worth mentioning that the dataset used in

this study contained only posed emotions with similar lighting and well-exposed faces. An in-the-wild image dataset, where the expressions are non-posed and the faces are at different angles, would definitely be more challenging to classify; therefore, it may require a different HOG configuration. Furthermore, to support the choice of `pixels_per_cell = (8, 8)` as the optimal value, statistical significance tests could be performed, comparing the classification scores of different PPC values.

Currently FER problems are mostly solved by deep learning architectures; however, classical feature extraction is still highly relevant for some purposes. Most notably, traditional machine learning algorithms often outperform deep neural networks when operating on limited datasets, such as the one utilized in this study. Deep learning models are inherently data-intensive and prone to overfitting when trained on small sample sizes. This research addresses a gap in the field of traditional classifiers using feature extraction, by evaluating the performance of different `pixels_per_cell` parameter values. However, future studies should investigate the remaining HOG variables, such as `orientations` and `cells_per_block`, to determine an optimal HOG configuration.

References

- [1] B. C. Ko, A brief review of facial emotion recognition based on visual information, *sensors* 18 (2018) 401.
- [2] A. Mollahosseini, B. Hasani, M. H. Mahoor, Affectnet: A database for facial expression, valence, and arousal computing in the wild, *IEEE transactions on affective computing* 10 (2017) 18–31.
- [3] L. Alzubaidi, J. Zhang, A. J. Humaidi, A. Al-Dujaili, Y. Duan, O. Al-Shamma, J. Santamaría, M. A. Fadhel, M. Al-Amidie, L. Farhan, Review of deep learning: concepts, cnn architectures, challenges, applications, future directions, *Journal of big Data* 8 (2021) 53.
- [4] Q. Wang, Advancements in gesture recognition: From traditional machine learning to deep learning innovations (2025).
- [5] N. Dalal, B. Triggs, Histograms of oriented gradients for human detection, in: 2005 IEEE computer society conference on computer vision and pattern recognition (CVPR'05), volume 1, Ieee, 2005, pp. 886–893.
- [6] O. Déniz, G. Bueno, J. Salido, F. De la Torre, Face recognition using histograms of oriented gradients, *Pattern recognition letters* 32 (2011) 1598–1603.
- [7] P. Carcagni, M. Del Coco, M. Leo, C. Distanti, Facial expression recognition and histograms of oriented gradients: a comprehensive study, *SpringerPlus* 4 (2015) 645.
- [8] P. Lucey, J. F. Cohn, T. Kanade, J. Saragih, Z. Ambadar, I. Matthews, The extended cohn-kanade dataset (ck+): A complete dataset for action unit and emotion-specified expression, in: 2010 IEEE computer society conference on computer vision and pattern recognition-workshops, IEEE, 2010, pp. 94–101.
- [9] S. Van der Walt, J. L. Schönberger, J. Nunez-Iglesias, F. Boulogne, J. D. Warner, N. Yager, E. Gouillart, T. Yu, scikit-image: image processing in python, *PeerJ* 2 (2014) e453.
- [10] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel,

- P. Prettenhofer, R. Weiss, V. Dubourg, et al., Scikit-learn: Machine learning in python, the Journal of machine Learning research 12 (2011) 2825–2830.
- [11] L. Breiman, Random forests, Machine learning 45 (2001) 5–32.
 - [12] M. Sokolova, G. Lapalme, A systematic analysis of performance measures for classification tasks, Information processing & management 45 (2009) 427–437.
 - [13] Dev-ShuvoAlok, Ck+ dataset, 2023. URL: <https://www.kaggle.com/datasets/shuvoalok/ck-dataset?>
 - [14] M. L. Smith, G. W. Cottrell, F. Gosselin, P. G. Schyns, Transmitting and decoding facial expressions, Psychological science 16 (2005) 184–189.